AI LAB 99

Machine Learning Track — Week 4, Module 7

Cluster Profiling & Business Segmentation Report

Turning a Two-Cluster Model into Business Segments, Personas, and Marketing Strategy

Dataset: Customer_Segmentation_Final.csv (scaled, clustered customer data)

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Prepared for AI Lab 99 | Customer Segmentation Curriculum

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Introduction & Dataset Overview

This report picks up after the clustering model was fit and profiles the resulting segments for business use: demographic and spending differences, shopping-channel behavior, marketing responsiveness, named business segments, customer personas, and final summary visuals.

Input file	Customer_Segmentation_Final.csv
Shape	2,038 rows × 42 columns
Feature values	Standardized (z-score) — see note below
Clusters found	2 (Cluster 0, Cluster 1)
Cluster sizes	Cluster 0: 1,111 customers (54.5%) · Cluster 1: 927 customers (45.5%)

Important note on units: every numeric column in this dataset (Income, Total_spending, purchase counts, etc.) was standardized with StandardScaler in the previous module and was never inverse-transformed back to real-world units for this analysis. All the “average” values reported below and throughout this notebook — e.g. “Income 0.67” — are standard-deviation units above or below the sample mean, not dollars or item counts. Relative comparisons between the two clusters are still valid (Cluster 0 is unambiguously higher-income and higher-spending than Cluster 1), but the numbers cannot be read as real income or spending figures without inverse-transforming them first. This report treats them as relative/directional throughout.

Activity 1 — Cluster Statistics & Demographic Analysis

Average feature values were computed per cluster and compared across demographics: age, income, education, marital status, family size, and number of children.

Metric (standardized)	Cluster 0 — Premium Loyal	Cluster 1 — Digital-First*
Customer count	1,111 (54.5%)	927 (45.5%)
Income	+0.67	−0.80
Total Spending	+0.81	−0.97
Customer Age	+0.18 (older)	−0.20 (younger)
Family Size	−0.32 (smaller)	+0.38 (larger)
Total Children	−0.37	+0.45
Web Purchases	+0.66	−0.78
Store Purchases	+0.71	−0.86
Catalog Purchases	+0.73	−0.88
Website Visits/Month	−0.40	+0.47
Deal Dependency	−0.60	+0.72

Metric (standardized)	Cluster 0 — Premium Loyal	Cluster 1 — Digital-First*
Recency (days since purchase)	+0.01	−0.02
Customer Activity Level	+0.71	−0.86

Cluster 0 skews slightly older, has fewer children and smaller households, but has meaningfully higher income. Cluster 1 skews slightly younger with larger families and more children, and meaningfully lower income.

Activity 2 — Spending Behavior Analysis

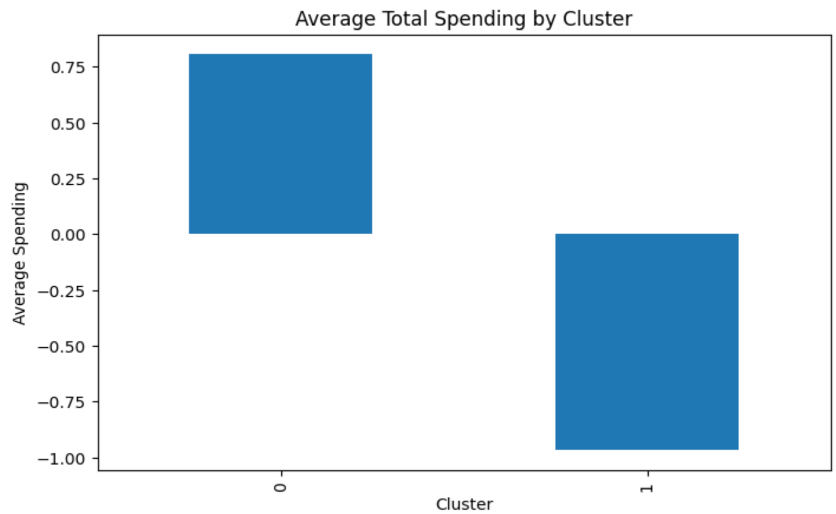

Figure 1. Average total spending by cluster (standardized units).

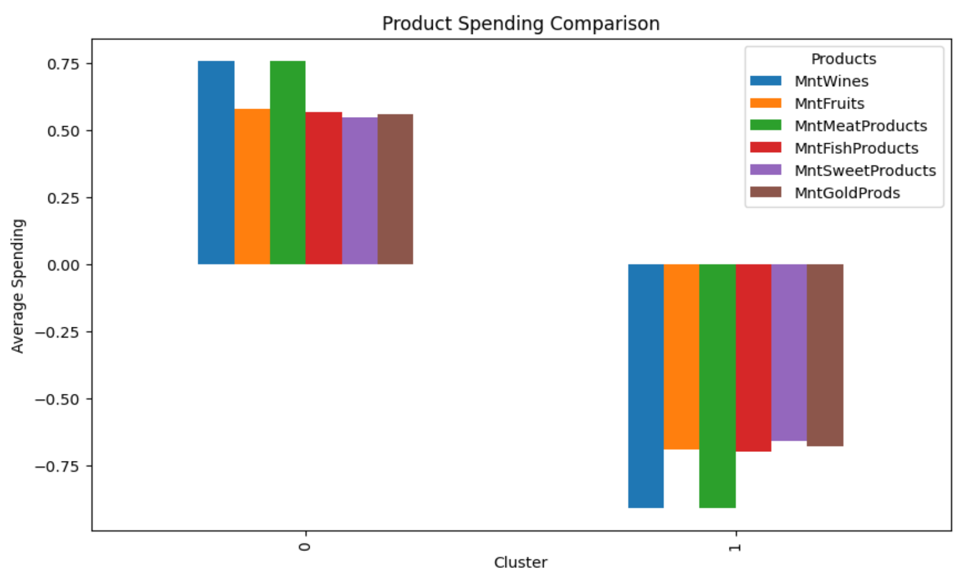

Figure 2. Product-category spending by cluster — every category is higher for Cluster 0.

Cluster 0 is both the highest-spending cluster (+0.81) and the premium-product cluster (highest combined wine + meat spending). Cluster 1 is the budget-conscious cluster (−0.97 total spending) on every single product category, with no exceptions — the two clusters separate cleanly on spending rather than specializing in different categories.

Activity 3 — Shopping Channel & Customer Engagement Analysis

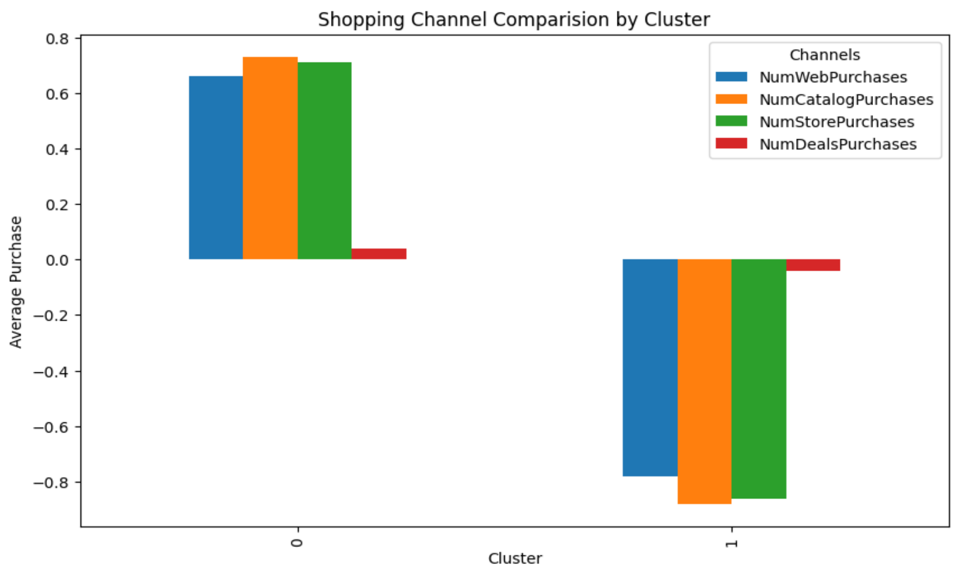

Figure 3. Purchases by channel (web, catalog, store, deals) per cluster.

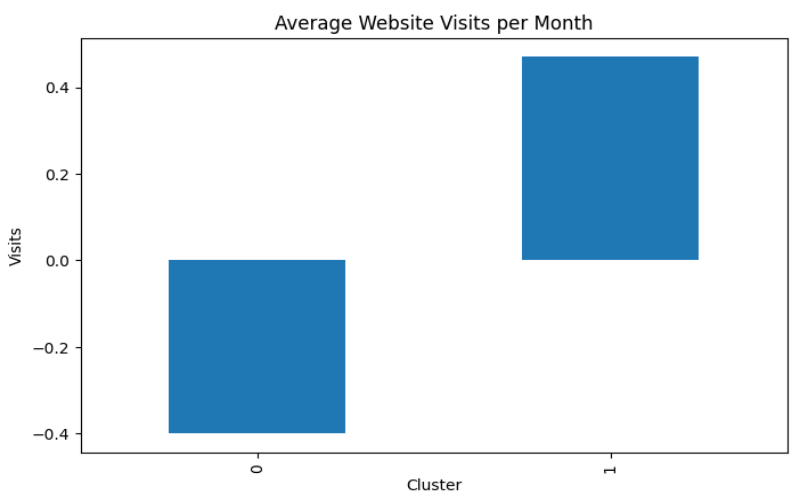

Figure 4. Average website visits per month by cluster.

Cluster 0 buys more through every channel — web, catalog, and store alike — so it is best described as the higher-engagement, higher-value cluster rather than a specific-channel cluster. Cluster 1 visits the website more often (+0.47) yet purchases less through every channel including web (−0.78); this is a browse-but-rarely-buy pattern rather than a genuinely "digital-first" purchasing pattern (see the naming caveat in

Activity 5). Cluster 1 also shows the higher deal-dependency score (+0.72 vs. −0.60), consistent with a more price-sensitive segment.

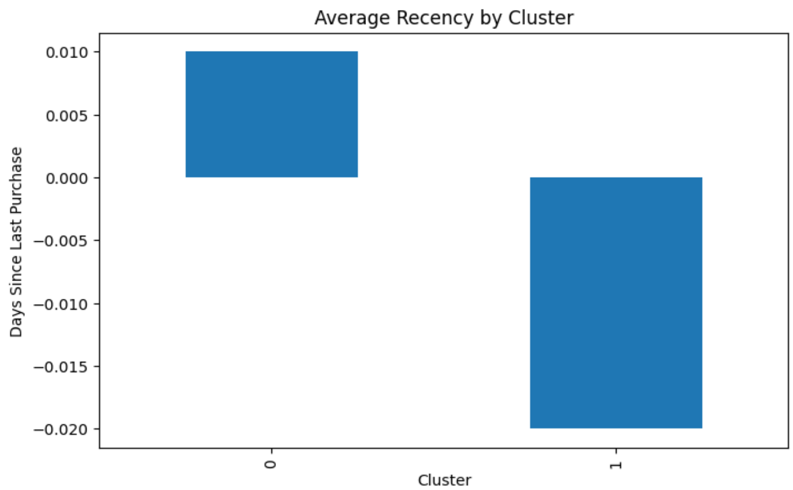

Figure 5. Average recency (days since last purchase) by cluster — nearly identical between clusters.

Activity 4 — Marketing Campaign Analysis

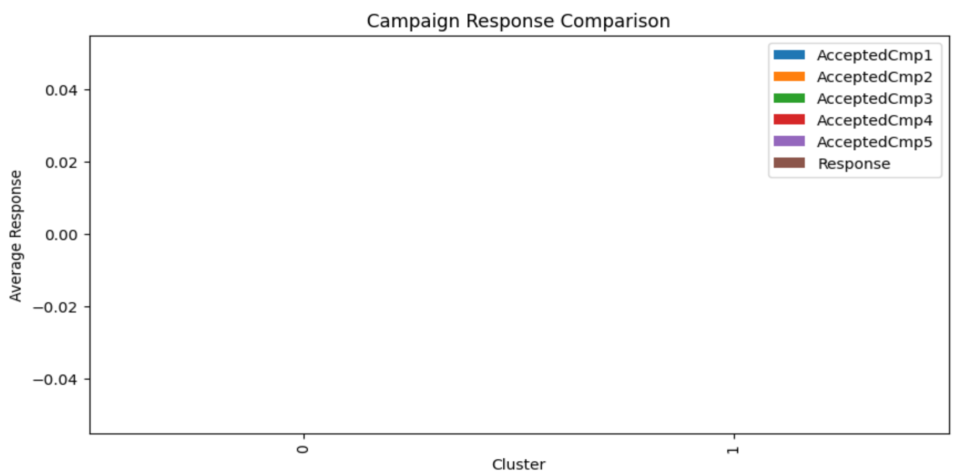

Figure 6. Campaign acceptance and response by cluster — flat at zero for both clusters and every campaign.

Data quality finding: AcceptedCmp1–AcceptedCmp5, Response, and Complain all average exactly 0.00 for both clusters, which makes this entire activity uninformative as delivered. Tracing this back to the Module 3 data-cleaning notebook, the root cause is IQR-based outlier capping (Winsorization) that was applied uniformly to every numeric column, including these near-constant binary flags. Because these columns are overwhelmingly 0 with a small minority of 1s, their IQR-derived upper bound sits at or near zero, so the capping step forced almost every 1 down to the bound — effectively deleting the signal before it ever reached this clustering stage. Binary/flag columns like these should be excluded from IQR-based outlier treatment in any future run of the pipeline.

Activity 5 — Business Segmentation & Cluster Naming

The two data clusters were mapped to business-friendly names, alongside two additional named segments that were not produced by the clustering model:

Segment	Persona	Age / Income	Shopping Habits	Data-Backed?
Premium Loyal Customers	Sarah	45–60 / High	Frequent premium purchases, store-preferred	**Yes — Cluster 0**
Digital-First Buyers	Ali	25–35 / Medium	Mostly online browsing; lowest purchases on every channel	**Yes — Cluster 1 (see naming caveat)**
Discount-Driven Families	Ahmed	30–45 / Medium	Buys during discounts	**No — no 3rd cluster exists**
At-Risk Customers	John	40–60 / Low	Rare purchases	**No — no 4th cluster exists**

Naming caveat: Cluster 1 was labeled “Digital-First Buyers,” but the data shows it has the lowest purchase volume on every channel, including web (−0.78) — the only channel where it leads Cluster 0 is website visits, not purchases. A more accurate label given the underlying numbers would be something like “Budget-Conscious Browsers” or “Low-Engagement Families.” The other two named segments — Discount-Driven Families and At-Risk Customers — do not correspond to any cluster in the 2-cluster solution and should be treated as aspirational personas for a possible future re-segmentation (e.g., a 4-cluster model), not as data-backed groups that exist in the current customer base.

Activity 6 — Customer Persona Development

Full personas (name, age range, income level, family status, shopping habits, preferred product/channel, marketing responsiveness, challenges, recommended strategy, estimated CLV) were built for all four named segments. As noted above, only the personas for Premium Loyal Customers (Sarah) and Digital-First Buyers (Ali) are grounded in actual customer data from this model; Ahmed and John are illustrative placeholders for segments the current 2-cluster model does not produce.

Sarah — Premium Loyal Customers	45–60, high income, small family, frequent premium (wine/meat) purchases in-store. VIP rewards & loyalty programs. Estimated CLV: Very High.
Ali — Digital-First Buyers	25–35, medium income, single/couple, browses online often but buys less than average on every channel. Needs personalized offers to convert visits into purchases. Estimated CLV: High.

Activity 7 — Cluster Visualization & Final Documentation

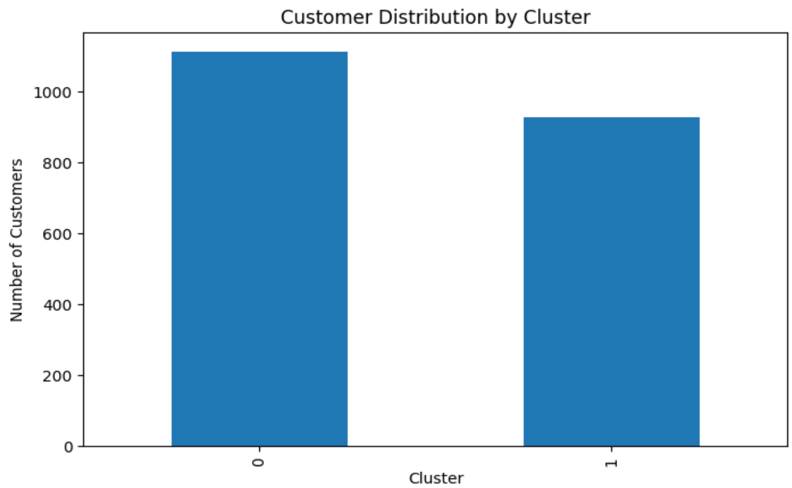

Figure 7. Customer count per cluster — 1,111 vs. 927, a reasonably balanced split.

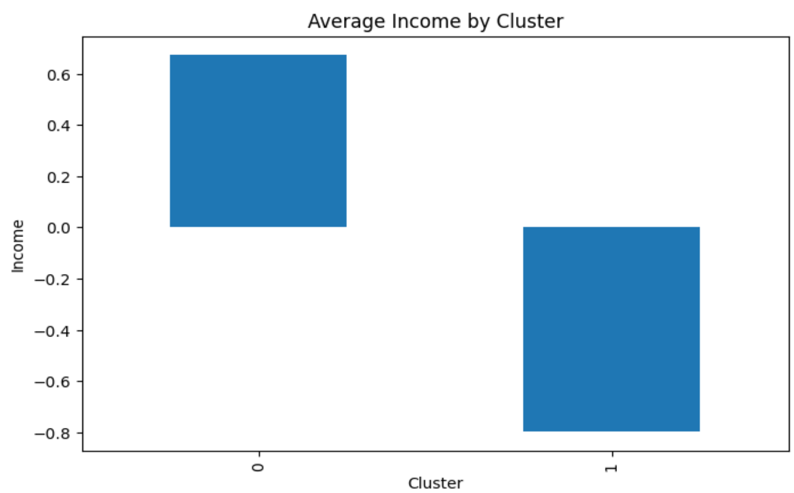

Figure 8. Average income by cluster.

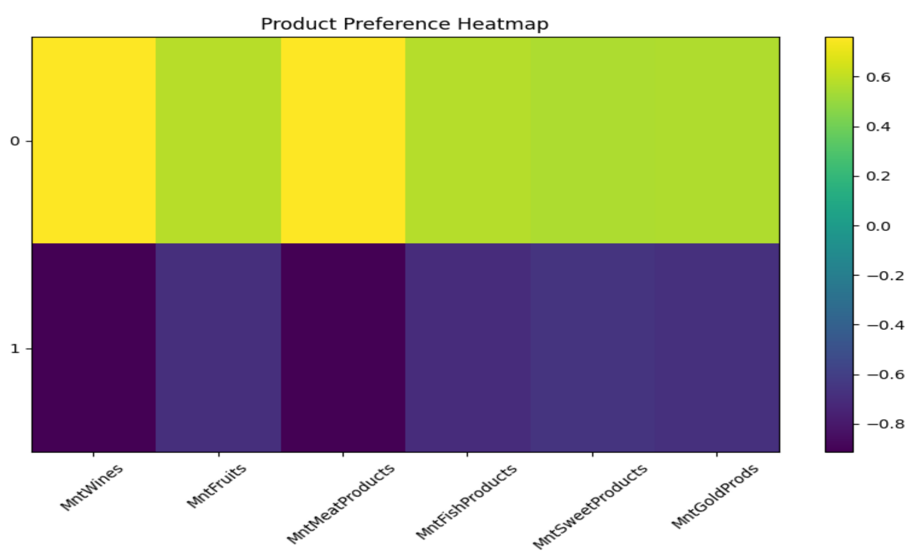

Figure 9. Product-preference heatmap — Cluster 0 (top row) outspends Cluster 1 on every product.

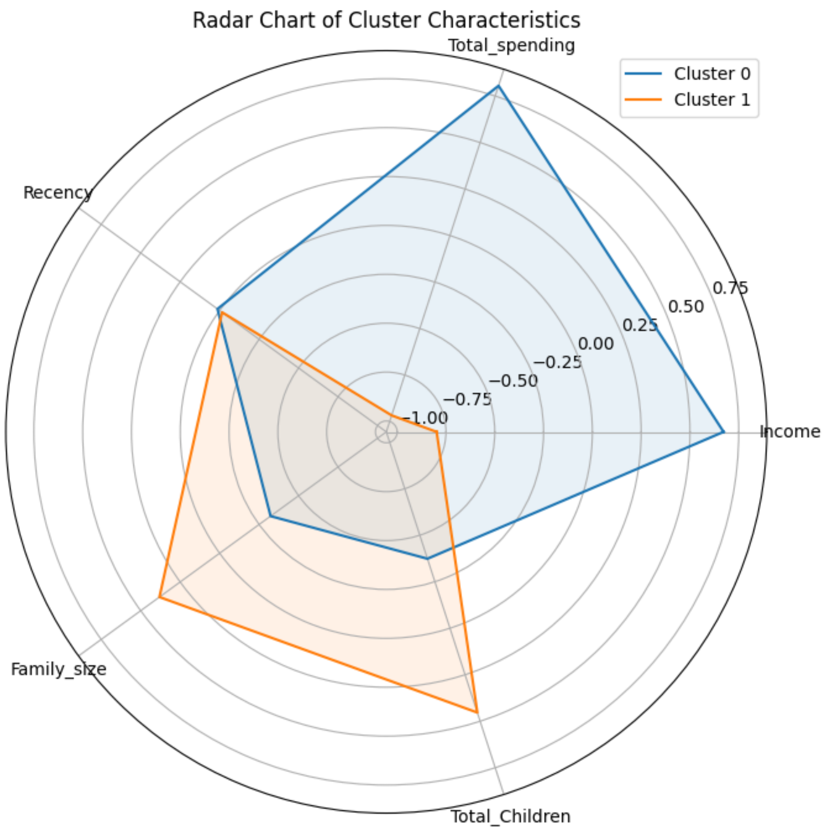

Figure 10. Radar chart summarizing income, spending, recency, family size, and children per cluster.

Two files were saved as final outputs: Segment_Comparison_Table.csv (the aggregated per-cluster metrics) and Final_Customer_Profile_Report.csv (the full 2,038-row labeled dataset).

Limitations & Recommendations

- **Standardized values not inverse-transformed:** every statistic in this report is a z-score, not a real-world unit. Before sharing these findings outside the modeling team, inverse-transform key metrics (Income, Total_spending) back to their original scale so stakeholders see dollar figures, not standard deviations.
- **Campaign and complaint signal was destroyed upstream:** re-run Module 3's outlier handling excluding binary/flag columns (AcceptedCmp1–5, Response, Complain, Z_CostContact, Z_Revenue) from IQR-based capping, then redo Activity 4 with real signal.
- **Segment count mismatch:** four business segments and personas were defined, but the model only produced two clusters. Either re-cluster with a higher k (e.g., k=4, informed by the earlier module's elbow/silhouette analysis) to genuinely produce Discount-Driven Families and At-Risk Customers, or drop those two segments from business-facing materials until they exist in the data.
- **“Digital-First Buyers” naming is inconsistent with the data:** Cluster 1 has the lowest purchase activity on every channel; consider renaming to reflect its actual browse-heavy, buy-light, price-sensitive pattern.
- **Only two clusters limits segmentation granularity:** a two-cluster split is a reasonable high-value/low-value starting point, but it cannot yet support the four distinct marketing strategies this

report's business tables describe. Validate additional cluster counts against silhouette/Davies-Bouldin scores before committing to four go-to-market personas.

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